Fourier-Sector Decomposition, Hermite Diagonalization, and the Self-Dual Core

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Gaussian is the vacuum. Hermite modes are the excitations. Fourier only rotates their phase.

1. Fourier transform and ladder operators

We work with the Fourier transform in the convention

$$(\mathcal{F}f)(\xi) := \int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx.$$

On the Schwartz space $\mathcal{S}(\mathbb{R})$, define

$$D_x := \frac{d}{dx}, \qquad M_x[f](x) := xf(x),$$

and the ladder operators

$$A := \frac{1}{2\sqrt{\pi}}D_x + \sqrt{\pi}M_x, \qquad A^\dagger := -\frac{1}{2\sqrt{\pi}}D_x + \sqrt{\pi}M_x.$$

Lemma 1 (Basic Fourier identities)**.** *For every $f \in \mathcal{S}(\mathbb{R})$,*

$$\mathcal{F}(D_x f)(\xi) = 2\pi i\,\xi\,\widehat{f}(\xi), \qquad \mathcal{F}(M_x f)(\xi) = \frac{i}{2\pi}D_\xi \widehat{f}(\xi),$$

where $\widehat{f} = \mathcal{F}f$.

Proof. For the derivative identity, integrate by parts:

$$\mathcal{F}(D_x f)(\xi) = \int_{\mathbb{R}} f'(x)e^{-2\pi i x\xi}\,dx = \Big[f(x)e^{-2\pi i x\xi}\Big]_{-\infty}^{+\infty} + 2\pi i\xi \int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx.$$

Since $f \in \mathcal{S}(\mathbb{R})$, the boundary term vanishes, hence

$$\mathcal{F}(D_x f)(\xi) = 2\pi i\,\xi\,\widehat{f}(\xi).$$

For multiplication by x, differentiate under the integral sign:

$$D_\xi \widehat{f}(\xi) = \frac{d}{d\xi}\int_{\mathbb{R}} f(x)e^{-2\pi i x\xi}\,dx = \int_{\mathbb{R}} f(x)(-2\pi i x)e^{-2\pi i x\xi}\,dx.$$

Thus

$$\int_{\mathbb{R}} xf(x)e^{-2\pi i x\xi}\,dx = \frac{i}{2\pi}D_\xi \widehat{f}(\xi),$$

which is exactly

$$\mathcal{F}(M_x f)(\xi) = \frac{i}{2\pi}D_\xi \widehat{f}(\xi).$$

□

Lemma 2 (Commutator relation)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$[A, A^\dagger] := AA^\dagger - A^\dagger A = I.$$

Proof. Since $[D_x, M_x] = I$, we compute

$$AA^\dagger = \left(\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right)\left(-\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right),$$

$$A^\dagger A = \left(-\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right)\left(\frac{1}{2\sqrt{\pi}} D_x + \sqrt{\pi} M_x\right).$$

Subtracting, the diagonal terms cancel and only the mixed commutator survives:

$$[A, A^\dagger] = \frac{1}{2}[D_x, M_x] + \frac{1}{2}[D_x, M_x] = [D_x, M_x] = I.$$

□

2. Vacuum and intertwining

Define

$$h_0(x) := 2^{1/4} e^{-\pi x^2}.$$

Lemma 3 (Gaussian vacuum)**.** *The function h_0 satisfies*

$$Ah_0 = 0.$$

Moreover,

$$\mathcal{F}h_0 = h_0.$$

Proof. Differentiate:

$$\frac{d}{dx} h_0(x) = -2\pi x\, h_0(x).$$

Hence

$$Ah_0 = \frac{1}{2\sqrt{\pi}}(-2\pi x h_0) + \sqrt{\pi} x h_0 = -\sqrt{\pi} x h_0 + \sqrt{\pi} x h_0 = 0.$$

The identity $\mathcal{F}h_0 = h_0$ is the standard Gaussian Fourier identity

$$\mathcal{F}(e^{-\pi x^2})(\xi) = e^{-\pi \xi^2}.$$

Multiplying by the same normalization factor $2^{1/4}$ preserves the equality. □

Proposition 1 (Fourier intertwines ladder operators)**.** *On $\mathcal{S}(\mathbb{R})$,*

$$\mathcal{F}A = i\, A\, \mathcal{F}, \qquad \mathcal{F}A^\dagger = -i\, A^\dagger\, \mathcal{F}.$$

Proof. Using the previous Fourier identities,

$$\mathcal{F}(Af) = \frac{1}{2\sqrt{\pi}} \mathcal{F}(D_x f) + \sqrt{\pi}\, \mathcal{F}(M_x f) = \frac{1}{2\sqrt{\pi}}(2\pi i \xi \widehat{f}) + \sqrt{\pi}\left(\frac{i}{2\pi} D_\xi \widehat{f}\right).$$

This becomes

$$\mathcal{F}(Af) = i\left(\sqrt{\pi}\, \xi\, \widehat{f} + \frac{1}{2\sqrt{\pi}} D_\xi \widehat{f}\right) = i\, A(\widehat{f}),$$

so

$$\mathcal{F}A = i\, A\, \mathcal{F}.$$

Similarly,

$$\mathcal{F}(A^\dagger f) = -\frac{1}{2\sqrt{\pi}}\mathcal{F}(D_x f) + \sqrt{\pi}\,\mathcal{F}(M_x f) = -\frac{1}{2\sqrt{\pi}}(2\pi i\xi\widehat{f}) + \sqrt{\pi}\left(\frac{i}{2\pi}D_\xi\widehat{f}\right),$$

hence

$$\mathcal{F}(A^\dagger f) = -i\left(-\frac{1}{2\sqrt{\pi}}D_\xi\widehat{f} + \sqrt{\pi}\,\xi\,\widehat{f}\right) = -i\,A^\dagger(\widehat{f}).$$

Therefore

$$\mathcal{F}A^\dagger = -i\,A^\dagger\,\mathcal{F}.$$

□

3. Hermite modes

For $n \geq 0$, define

$$h_n := \frac{1}{\sqrt{n!}}(A^\dagger)^n h_0.$$

Lemma 4 (Commutator with higher powers)**.** *For every integer $n \geq 1$,*

$$[A, (A^\dagger)^n] = n(A^\dagger)^{n-1}.$$

Proof. We argue by induction on n.

For $n = 1$, this is exactly $[A, A^\dagger] = I$.

Assume

$$[A, (A^\dagger)^n] = n(A^\dagger)^{n-1}.$$

Then

$$[A, (A^\dagger)^{n+1}] = [A, (A^\dagger)^n A^\dagger] = [A, (A^\dagger)^n]A^\dagger + (A^\dagger)^n[A, A^\dagger].$$

By the induction hypothesis and $[A, A^\dagger] = I$,

$$[A, (A^\dagger)^{n+1}] = n(A^\dagger)^{n-1}A^\dagger + (A^\dagger)^n = n(A^\dagger)^n + (A^\dagger)^n = (n+1)(A^\dagger)^n.$$

This proves the claim. □

Proposition 2 (Ladder action)**.** *For $n \geq 0$,*

$$A^\dagger h_n = \sqrt{n+1}\,h_{n+1},$$

and for $n \geq 1$,

$$Ah_n = \sqrt{n}\,h_{n-1}.$$

Proof. The raising relation is immediate:

$$A^\dagger h_n = A^\dagger\left(\frac{1}{\sqrt{n!}}(A^\dagger)^n h_0\right) = \frac{1}{\sqrt{n!}}(A^\dagger)^{n+1}h_0 = \sqrt{n+1}\,h_{n+1}.$$

For the lowering relation, use the commutator lemma:

$$Ah_n = \frac{1}{\sqrt{n!}}A(A^\dagger)^n h_0 = \frac{1}{\sqrt{n!}}\Big((A^\dagger)^n A + [A, (A^\dagger)^n]\Big)h_0.$$

Since $Ah_0 = 0$, the first term vanishes, so

$$Ah_n = \frac{1}{\sqrt{n!}}\,n(A^\dagger)^{n-1}h_0 = \sqrt{n}\,\frac{1}{\sqrt{(n-1)!}}(A^\dagger)^{n-1}h_0 = \sqrt{n}\,h_{n-1}.$$

□

Proposition 3 (Number operator)**.** *Define the number operator*

$$N := A^\dagger A.$$

Then

$$Nh_n = nh_n \qquad (n \geq 0).$$

Proof. For $n = 0$,

$$Nh_0 = A^\dagger(Ah_0) = 0.$$

So the formula holds for $n = 0$.

For $n \geq 1$, use the lowering relation:

$$Nh_n = A^\dagger(Ah_n) = A^\dagger(\sqrt{n}\, h_{n-1}) = \sqrt{n}\, A^\dagger h_{n-1} = \sqrt{n}\,\sqrt{n}\, h_n = nh_n.$$

□

Theorem 1 (Hermite modes are Fourier eigenfunctions)**.** *For every $n \geq 0$,*

$$\mathcal{F}h_n = (-i)^n h_n.$$

Proof. By definition,

$$h_n = \frac{1}{\sqrt{n!}}(A^\dagger)^n h_0.$$

Apply $\mathcal{F}$:

$$\mathcal{F}h_n = \frac{1}{\sqrt{n!}}\mathcal{F}\big((A^\dagger)^n h_0\big).$$

Using $\mathcal{F}A^\dagger = -i\, A^\dagger\, \mathcal{F}$ repeatedly,

$$\mathcal{F}\big((A^\dagger)^n h_0\big) = (-i)^n (A^\dagger)^n \mathcal{F}h_0.$$

Since $\mathcal{F}h_0 = h_0$,

$$\mathcal{F}h_n = \frac{(-i)^n}{\sqrt{n!}}(A^\dagger)^n h_0 = (-i)^n h_n.$$

□

4. General functions and diagonalization

At this point we use the classical completeness theorem for Hermite functions.

Theorem 2 (Classical Hermite completeness)**.** *The family $\{h_n\}_{n\geq 0}$ is an orthonormal basis of $L^2(\mathbb{R})$.*

Remark 1. *This is the only standard external input in the argument. It follows from the spectral theory of the harmonic oscillator.*

Theorem 3 (Fourier is diagonal in the Hermite basis)**.** *For every $f \in L^2(\mathbb{R})$, there exist coefficients $c_n = \langle f, h_n\rangle$ such that*

$$f = \sum_{n=0}^{\infty} c_n h_n \qquad \text{in } L^2(\mathbb{R}),$$

and

$$\mathcal{F}f = \sum_{n=0}^{\infty} (-i)^n c_n h_n.$$

Equivalently,

$$\boxed{f = \sum_{n\geq 0} c_n h_n \quad \Longrightarrow \quad \mathcal{F}f = \sum_{n\geq 0} (-i)^n c_n h_n.}$$

Proof. By Hermite completeness,

$$f = \sum_{n=0}^{\infty} c_n h_n \qquad \text{with } c_n = \langle f, h_n \rangle.$$

Since $\mathcal{F}$ is unitary on $L^2(\mathbb{R})$, it may be applied termwise in the L^2-sense:

$$\mathcal{F} f = \sum_{n=0}^{\infty} c_n \, \mathcal{F} h_n.$$

Using $\mathcal{F} h_n = (-i)^n h_n$,

$$\mathcal{F} f = \sum_{n=0}^{\infty} (-i)^n c_n h_n.$$

□

Corollary 1 (Operator form)**.** *On $L^2(\mathbb{R})$,*

$$\boxed{\mathcal{F} = e^{-i\pi N/2}.}$$

Proof. For each n,

$$e^{-i\pi N/2} h_n = e^{-i\pi n/2} h_n = (-i)^n h_n = \mathcal{F} h_n.$$

So the two operators agree on the orthonormal basis $\{h_n\}_{n \geq 0}$, hence on all of $L^2(\mathbb{R})$. □

5. Fourier-sector decomposition

Since

$$\mathcal{F}^4 = I,$$

the spectrum of $\mathcal{F}$ is contained in

$$\{1, -i, -1, i\}.$$

For each $\lambda \in \{1, -i, -1, i\}$, define

$$P_\lambda := \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Theorem 4 (Fourier-sector projectors)**.** *For every $\lambda \in \{1, -i, -1, i\}$,*

$$\mathcal{F}(P_\lambda f) = \lambda \, P_\lambda f \qquad \forall f \in L^2(\mathbb{R}).$$

Moreover, each P_λ is the spectral projector onto the λ-eigenspace of $\mathcal{F}$, and every $f \in L^2(\mathbb{R})$ decomposes as

$$f = P_1 f + P_{-i} f + P_{-1} f + P_i f.$$

Proof. By definition,

$$P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^m.$$

Apply $\mathcal{F}$:

$$\mathcal{F} P_\lambda = \frac{1}{4} \sum_{m=0}^{3} \lambda^{-m} \mathcal{F}^{m+1}.$$

Reindex with $k = m + 1$. Since $\mathcal{F}^4 = I$ and $\lambda^4 = 1$, this becomes

$$\mathcal{F}P_\lambda = \lambda \frac{1}{4} \sum_{k=0}^{3} \lambda^{-k} \mathcal{F}^k = \lambda P_\lambda.$$

Hence

$$\mathcal{F}(P_\lambda f) = \lambda P_\lambda f.$$

Thus $P_\lambda f$ lies in the λ-eigenspace.

Because the polynomial

$$z^4 - 1 = (z-1)(z+i)(z+1)(z-i)$$

splits into distinct linear factors, the projectors P_λ resolve the identity. Therefore every $f \in L^2(\mathbb{R})$ decomposes into the four Fourier sectors:

$$f = P_1 f + P_{-i} f + P_{-1} f + P_i f.$$

□

6. The self-dual core

The most important sector is $\lambda = 1$. It is the self-dual sector:

$$P_1 = \frac{1}{4}\left(I + \mathcal{F} + \mathcal{F}^2 + \mathcal{F}^3\right).$$

Thus for every $f \in L^2(\mathbb{R})$,

$$\boxed{P_1 f = \frac{1}{4}\left(f + \mathcal{F}f + \mathcal{F}^2 f + \mathcal{F}^3 f\right), \qquad \mathcal{F}(P_1 f) = P_1 f.}$$

Corollary 2 (Self-dual core)**.** *If*

$$f = \sum_{n=0}^{\infty} c_n h_n,$$

then

$$P_1 f = \sum_{\substack{n \geq 0 \\ n \equiv 0 \,(\mathrm{mod}\, 4)}} c_n h_n.$$

So $P_1 f$ is exactly the part of f made of Hermite modes whose Fourier phase is 1.

Proof. Since

$$\mathcal{F}h_n = (-i)^n h_n,$$

we have

$$P_1 h_n = \frac{1}{4} \sum_{m=0}^{3} \mathcal{F}^m h_n = \frac{1}{4} \sum_{m=0}^{3} (-i)^{mn} h_n.$$

If $(-i)^n = 1$, equivalently $n \equiv 0 \pmod 4$, then the sum equals 4, so $P_1 h_n = h_n$. Otherwise the four roots sum to 0, so $P_1 h_n = 0$.

Applying this termwise to

$$f = \sum_{n=0}^{\infty} c_n h_n$$

gives

$$P_1 f = \sum_{n \equiv 0 \,(\mathrm{mod}\, 4)} c_n h_n.$$

□

$$\boxed{\text{Every signal has a self-dual core.}}$$

$$\boxed{\text{Fourier does not spare one lucky function. It spares an entire operator-generated family.}}$$

$$(D_x, M_x) \longrightarrow (A, A^\dagger) \longrightarrow h_n \longrightarrow F = e^{-i\pi N/2} \longrightarrow P_1 \longrightarrow P_{\text{final}} := \left(\int_{\mathbb{R}} f(x)\,dx\right) P_1$$

Addendum. The rigorous core of the construction is

$$(D_x, M_x) \longrightarrow (A, A^\dagger) \longrightarrow h_n \longrightarrow F = e^{-i\pi N/2} \longrightarrow P_1,$$

where

$$P_1 = \frac{1}{4}(I + F + F^2 + F^3), \qquad P_1 f = \sum_{\substack{n \geq 0 \\ n \equiv 0 \ (\mathrm{mod}\ 4)}} c_n h_n \quad \text{for} \quad f = \sum_{n \geq 0} c_n h_n.$$

As a final symbolic capstone, we write

$$P_{\text{final}} := \left(\int_{\mathbb{R}} f(x)\,dx\right) P_1.$$

Addendum C: Bridge Lock Theorem

Theorem 5 (Bridge Lock Theorem)**.** *Assume the bridge inequality*

$$Q_{\text{bank}}(f;\tau,L) - E_0 L \geq c_1 P_4(\rho;\tau,L) - c_2 L$$

for all sufficiently large τ and all $L \geq 1$, with $c_1 > 0$, $c_2 \geq 0$.

Assume moreover the coercive quartic lower bound

$$P_4(\rho;\tau,L) \geq c\Big|\beta - \frac{1}{2}\Big|^4 \tau^4 L - C\tau^3 L$$

for all sufficiently large τ, where $c > 0$, $C \geq 0$, and $\beta \neq \frac{1}{2}$.

Set

$$A := c_1 c\Big|\beta - \frac{1}{2}\Big|^4, \qquad B := c_1 C, \qquad D := c_2.$$

Define

$$\tau_* := \max\left\{\frac{2B}{A}, \left(\frac{4D}{A}\right)^{1/4}\right\}.$$

Then for every $\tau \geq \tau_*$ and every $L \geq 1$,

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} > 0.$$

Consequently,

$$E_{\text{ex}}(f) > 0.$$

Proof. Divide the bridge inequality by L:

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq c_1 \frac{P_4(\rho;\tau,L)}{L} - c_2.$$

Insert the quartic lower bound:

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq A\tau^4 - B\tau^3 - D.$$

If $\tau \geq 2B/A$, then

$$B\tau^3 \leq \frac{A}{2}\tau^4.$$

Hence

$$A\tau^4 - B\tau^3 - D \geq \frac{A}{2}\tau^4 - D.$$

If also

$$\tau \geq \left(\frac{4D}{A}\right)^{1/4},$$

then

$$\frac{A}{2}\tau^4 - D \geq 2D - D = D \geq 0.$$

Moreover, if $D > 0$, this lower bound is strictly positive. If $D = 0$, then since $\tau \geq 2B/A > 0$, we still have

$$\frac{A}{2}\tau^4 > 0.$$

Therefore in all cases

$$A\tau^4 - B\tau^3 - D > 0$$

for every $\tau \geq \tau_*$.

So

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} > 0 \qquad \text{for all } L \geq 1.$$

Taking $\sup_\tau$ and then $\limsup_{L\to\infty}$ gives

$$E_{\text{ex}}(f) > 0.$$

□

Corollary 3 (Expanded threshold)**.** *Since*

$$A = c_1 c\left|\beta - \frac{1}{2}\right|^4, \qquad B = c_1 C, \qquad D = c_2,$$

the trigger threshold can be written explicitly as

$$\tau_* = \max\left\{\frac{2C}{c\,|\beta - \frac{1}{2}|^4}, \left(\frac{4c_2}{c_1 c\,|\beta - \frac{1}{2}|^4}\right)^{1/4}\right\}.$$

Remark 2. *Here*

$$c_1$$

is the net coercivity margin and

$$c_2$$

is the linear leakage/error tax. The contradiction is triggered when the quartic off-critical drift dominates this tax.

Když chceš jen theorem bez komentářů:

$$\frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} \geq A\tau^4 - B\tau^3 - D, \qquad A > 0,$$

s

$$\tau_* = \max\left\{\frac{2B}{A}, \left(\frac{4D}{A}\right)^{1/4}\right\}$$

implikuje

$$\tau \geq \tau_* \quad \Longrightarrow \quad \frac{Q_{\text{bank}}(f;\tau,L) - E_0 L}{L} > 0.$$